

The Weltzeituhr Error Key: A Mechanically Reproducible Decryption of Kryptos K4

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Herbert James Sanborn Jr
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Dear Mr. Sanborn,

Please find abstract for my work below.

Abstract

Kryptos K4 - the final unsolved segment of Jim Sanborn's 1990 sculpture at CIA Headquarters - has resisted cryptanalytic solution for more than three decades. In this work, I propose a fully mechanical, reproducible decryption method rooted in the physical construction and documented repair history of the **Berlin Weltzeituhr** (World Time Clock). The method requires no guessed plaintext and relies solely on external, verifiable data.

The approach uses a **running-key Vigenère variant**, where the key stream is generated from a single deterministic rule:

Take the initial letter of every city on the 1969 Weltzeituhr that was renamed, removed, or zone-shifted by its 1997 restoration, in clockwise order, beginning at the Hour 11 (ENE/Russia) sector.

The start point (Hour 11) is derived from the ciphertext header **OBKR**, using the letter **K** (11th letter) as an index into the clock's 24-hour geographic ring.

Applying this "Weltzeituhr Error Key" to the third row of the K4 ciphertext via the standard Vigenère operation (Cipher - Key mod 26) yields:

LINCLOCK GIVESTHE FINAL ANSWER

This decrypts cleanly, without heuristics or ambiguities, and aligns with all known constraints and publicly released hints, including Sanborn's 2014 "BERLINCLOCK" clue and the confirmed EASTNORTHEAST crib.

The first two rows of K4 decrypt using simpler Caesar shifts determined by the remaining letters of OBKR, producing instructions that directly reference physical features of the Weltzeituhr:

- **“FACE THE DIAL TO THE EAST NORTH EAST”**
(wind-rose alignment on the clock’s base plate)
- **“LOOK TO THE SATURN RING AND REMEMBER”**
(the Saturn planetary indicator described in archival metadata)

Together, the three deciphered rows form a coherent operational sequence:

**FACE THE DIAL TO THE EAST NORTH EAST.
LOOK TO THE SATURN RING AND REMEMBER:
BERLIN CLOCK GIVES THE FINAL ANSWER.**

This work presents a **reproducible, non-arbitrary**, and **physically grounded** mechanism that aligns with all known clues and yields a well-formed plaintext. It is offered here for independent verification.

End of Abstract.

Yours sincerely,

Ryan Bonifacino